Data Contracts

The three tracks (Digitization, Model, Frontend) work in parallel. They only touch each other through the files and API below. **Change a contract only by agreement of all three tracks**, and bump the `version` line when you do.

`version: 1 (2026-10-10)`



1. `data/sjcet/sjcet_daily.csv` — the digitized register (Digitization → Model, Frontend)

One row per calendar day, **including days with no reading** (all value columns empty).

column	type	unit	notes
`date`	`YYYY-MM-DD`	–	the date written in the register
`obs_time`	`HH:MM`	IST	time of reading, empty if not written
`rain_mm`	float	mm	**register is in cm → multiply by 10**. `0` = "Nil"/"–" written; empty = not recorded
`dry_bulb_c`	float	°C	
`wet_bulb_c`	float	°C	
`tmin_c`	float	°C	
`tmax_c`	float	°C	
`rh_pct`	float	%	derived from dry/wet bulb (see digitization guide §11), never OCR'd
`source_page`	string	–	`page_id` from `pages.csv`
`qc_flag`	enum	–	`ok` · `corrected` (human fixed) · `suspect` (kept but doubtful)
`qc_note`	string	–	free text, optional

Rules:

- Missing ≠ zero. An empty cell means "not observed"; `0` means "observed, no rain".
- 8:30 AM register convention: `rain_mm` on date D is rain from D-1 08:30 to D 08:30; `tmax_c` on date D is normally the max of day D-1. The model track handles the shift; the CSV stores values **exactly against the date written**.

2. `data/sjcet/cells.csv` — every OCR'd cell (Digitization → Frontend review tool)

column	example	notes
`cell_id`	`p0123_r05_c3`	unique, stable
`page_id`	`p0123`	
`page_path`	`digitization/work/pages/p0123.png`	relative to repo root
`crop_path`	`digitization/work/cells/p0123_r05_c3.png`	
`bbox`	`412,980,530,1032`	`x0,y0,x1,y1` in page pixels

column	example	notes
`date`	`2012-07-05`	from row → day mapping
`field`	`tmax_c`	one of the value columns above, or `date`
`ocr_text`	`29.5`	raw model output
`ocr_conf`	`0.83`	0–1
`value`	`29.5`	parsed number (empty if unparseable)
`qc_flag`	`suspect`	`ok` / `suspect` / `fail`
`qc_reason`	`tmax<tmin`	short machine reason

3. `data/sjcet/corrections.csv` — human review results (Review tool → Digitization)

Append-only. The latest row per `cell_id` wins.

column	notes
`cell_id`	
`value`	corrected text as it should be read, e.g. `29.5`
`status`	`accepted` (OCR was right) · `corrected` · `illegible` · `blank`
`reviewer`	initials
`timestamp`	ISO 8601

4. Forecast JSON (Model → Frontend)

Served at `GET /api/forecast?date=YYYY-MM-DD`, also written to `model/outputs/forecasts/<date>.json`. `issued` is the last day of input data.

```
{
  "version": 1,
  "issued": "2015-07-10",
  "station": {"id": "SJCET", "name": "SJCET Palai (Choondacherry)", "lat": 9.727, "lon": 76.727},
  "model": {"name": "gnn-stage3", "run_id": "2026-11-02_a"},
  "days": [
    {"date": "2015-07-11", "tmax_c": 29.1, "tmin_c": 22.8, "tmean_c": 26.0, "dewpoint_c": 23.4,
     "rh_pct": 86, "rain_mm": 12.4, "rain_prob": 0.71, "rain_expected": true,
     "observed": {"tmax_c": 28.6, "tmin_c": 23.0, "tmean_c": 25.7, "dewpoint_c": 23.1, "rh_pct": 87,
"rain_mm": 9.0}}
  ]
}
```

`observed` is present only when the date is in the past (replay mode) and data exist; any value may be `null`. `rh_pct` (relative humidity) is calculated from `tmean_c` and `dewpoint_c`.

5. Metrics JSON (Model → Frontend)

`model/outputs/metrics.json`:

```
{
  "version": 1,
  "test_period": ["2015-01-01", "2016-12-31"],
  "models": [
    {"name": "persistence", "group": "baseline",
     "metrics": {"tmax_c": {"rmse": 1.9, "mae": 1.4, "bias": 0.0, "r2": 0.31},
                 "tmin_c": {"rmse": 1.2, "mae": 0.9, "bias": 0.0, "r2": 0.40},
                 "rain":   {"rmse": 14.2, "mae": 7.1, "hit_rate": 0.62, "false_alarm": 0.35},
                 "rh_pct": {"rmse": 3.8, "mae": 2.9, "bias": 0.1, "r2": 0.5}}}
  ],
  "ablation": [
    {"model": "gnn-1-2-3", "local_fraction": 0.25, "seed": 0, "tmax_rmse": 1.3, "tmin_rmse": 0.9,
"rain_rmse": 12.1}
  ],
  "by_season": [
    {"model": "gnn-1-2-3", "season": "SW monsoon (Jun–Sep)", "tmax_rmse": 1.1}
  ]
}
```


Seasons (IMD): Winter Jan–Feb · Pre-monsoon Mar–May · SW monsoon Jun–Sep · NE monsoon Oct–Dec.

6. Graph JSON (Model → Frontend map)

`model/outputs/graph.json`:


```
{"nodes": [{"id": 0, "kind": "era5", "lat": 9.75, "lon": 76.75},
           {"id": 401, "kind": "gsod", "lat": 8.48, "lon": 76.95, "name": "THIRUVANANTHAPURAM"},
           {"id": 402, "kind": "sjcet", "lat": 9.727, "lon": 76.727}],
 "edges": [[0, 402], [401, 0]]}
```


`kind` ∈ `era5` · `gsod` · `sjcet`.